



Name.....

Teacher.....

2012
Preliminary Course
FINAL EXAMINATION

Mathematics

General Instructions

- Working Time - 2 hours.
- Write using a blue or black pen.
- Approved calculators may be used.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total marks – 90

Section I Pages 3–4 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

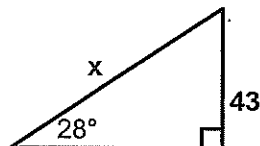
Section II Pages 5–12 80 marks

Attempt Questions 11–16
Allow about 1 hours 45 minutes for this section

1. The number 147.658 correct to two significant figures is

(a) 15 (b) 150 (c) 147.65 (d) 147.66

2. The value of x is given by



(a) $43 \times \cos 28^\circ$ (b) $43 \times \sin 28^\circ$

(c) $\frac{43}{\cos 28^\circ}$ (d) $\frac{43}{\sin 28^\circ}$

3. If $4y^2 - 12y + P = (2y + Q)^2$ then

(a) $P = 9$ $Q = 3$ (b) $P = 9$ $Q = -3$

(c) $P = -9$ $Q = -3$ (d) $P = -9$ $Q = 3$

4. The gradient of any line perpendicular to the line $3x - 2y + 12 = 0$ is

(a) $\frac{3}{2}$ (b) $-\frac{3}{2}$ (c) $\frac{2}{3}$ (d) $-\frac{2}{3}$

5. $\frac{2}{\sqrt{2}} + \frac{\sqrt{2}}{2} =$

(a) 1 (b) 2 (c) $\frac{3\sqrt{2}}{2}$ (d) $\frac{2 + \sqrt{2}}{2\sqrt{2}}$

6. Simplify $\frac{x^2 - 1}{(x - 1)^2}$

(a) 1 (b) $\frac{-1}{-2x + 1}$ (c) $\frac{x + 1}{x - 1}$ (d) $\frac{x - 1}{x + 1}$

Question 11 (10 Marks)

Use a Separate Sheet of paper

Marks

- (a) Find the value of $\frac{4.23}{\sqrt{6.14 - 1.78}}$, giving your answer correct to 2 decimal places. **1**
- (b) If $s = \frac{a}{1-r}$ find s when $a = 7, r = \frac{1}{3}$. **1**
- (c) If $\sqrt{12} + \sqrt{27} = \sqrt{a}$, find the value of a . **2**
- (d) Express $\frac{2}{5-2\sqrt{5}}$ as a fraction with a rational denominator. **2**
- (e) Fully factorise the following expressions fully
- (i) $16x^3y - 2y^4$ **2**
- (ii) $mx^2 + my^2 - nx^2 - ny^2$ **2**

Question 13 (10 Marks)

Use a Separate Sheet of paper

Marks

(a) A function is defined by the rule $g(x) = \begin{cases} x+1, & \text{if } x \geq 1 \\ -1, & \text{if } -2 < x < 1 \\ 1-x, & \text{if } x \leq -2 \end{cases}$

Find

(i) $g(1)$

1

(ii) $g(-3) + g(0)$

1

(b) Sketch the graphs of the following, **stating the domain and range of each.**

(i) $y = \frac{2}{x}$

2

(ii) $x^2 + y^2 = 25$

2

(iii) $3(x+2) - y = 0$

2

(c) Show that the function $f(x) = \frac{1-x^2}{x}$ is an odd function.

2

Question 15 (10 Marks)

Use a Separate Sheet of paper

Marks

The points $A(2,0)$, $B(8,4)$, $C(4,6)$ and $D(x_1, y_1)$ form the 4 vertices of a parallelogram ABCD.

- | | | |
|-----|---|---|
| (a) | Draw a number plane and plot the points A , B & C .
(USE A RULER) | 1 |
| (b) | Find the gradient of line AB | 1 |
| (c) | Show that the equation of the line l parallel to AB and passing through C is $2x - 3y + 10 = 0$ | 2 |
| (d) | Find the point $D(x_1, y_1)$ and mark this point on your diagram. | 2 |
| (e) | Find the angle θ to the nearest degree that the line AB makes with the positive x -axis | 2 |
| (f) | Find the perpendicular distance between the line l and A . | 2 |

Question 17 (10 Marks)

Use a Separate Sheet of paper

Marks

- (a) Find the derivative of the following: (You do not need to simplify your answers after finding the derivative.)

(i) $x^4 - 3x^3 + 2$

1

(ii) $\sqrt{x^3}$

1

(iii) $\frac{1}{3x^4}$

1

- (b) Find $f'(2)$ for $f(x) = (3x^2 - 5x)^5$.

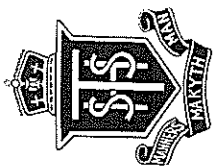
2

- (c) Given $y = \frac{x^2 - 1}{x^2 + 1}$ find $\frac{dy}{dx}$

2

- (d) Find the equation of the normal to the curve $y = 2x^3 - 4x^2$ at the point (1, -2)

3



Sydney Technical High School

Preliminary Examination
Mathematics 2012

Multiple Choice Answer Sheet

Name _____

Teacher _____

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☒ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☒
3. A ☐ B ☒ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☐ C ☒ D ☐
6. A ☐ B ☐ C ☒ D ☐
7. A ☒ B ☐ C ☐ D ☐
8. A ☐ B ☒ C ☐ D ☐
9. A ☐ B ☒ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☒

Question 11

a) 2.03 (2dp)

b) $S = \frac{a}{1-r}$

$S = \frac{7}{1-\frac{1}{3}}$

$S = 10\frac{1}{2}$

c) $\sqrt{12} + \sqrt{27} = \sqrt{a}$
 $2\sqrt{3} + 3\sqrt{3} = \sqrt{a}$
 $5\sqrt{3} = \sqrt{a}$
 $\sqrt{25 \times 3} = \sqrt{a}$
 $a = 75$

d) $\frac{2}{5-2\sqrt{5}} \times \frac{5+2\sqrt{5}}{5+2\sqrt{5}}$

$= \frac{10 + 4\sqrt{5}}{25 - 20}$

$= \frac{10 + 4\sqrt{5}}{5}$

e) i) $2y(8x^3 - y^3) = 2y(2x - y)(4x^2 + 2xy + y^2)$

ii) $mx^2 + my^2 - nx^2 - ny^2 = m(x^2 + y^2) - n(x^2 + y^2)$
 $= (m - n)(x^2 + y^2)$

Question 14

a) $\cos 135^\circ = \frac{-1}{\sqrt{2}}$ or $-\frac{\sqrt{2}}{2}$

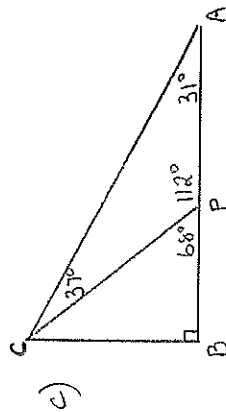
ii) $\cot 210^\circ = \sqrt{3}$

b) $2\sin^2\theta - 1 = 0$

$\sin^2\theta = \frac{1}{2}$

$\sin\theta = \pm\sqrt{\frac{1}{2}}$

$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$



$$\frac{CP}{\sin 31^\circ} = \frac{20}{\sin 37^\circ}$$

$$CP = \frac{20 \times \sin 31^\circ}{\sin 37^\circ}$$

$$CP \approx 17.12$$

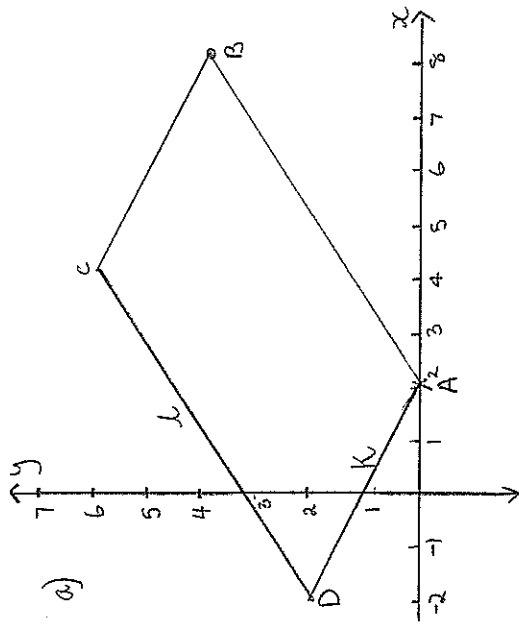
ciii) $\cos 68^\circ = \frac{BP}{17.12}$

$$BP = 17.12 \cos 68^\circ$$

$$BP \approx 6.41$$

$$\approx 6 \text{ cm}$$

Question 15



d) i) $\sin C = \frac{2}{3}$

$$\text{Area} = \frac{1}{2}ab \sin C$$

$$= \frac{1}{2} \times 9.8 \times 12.6 \times \frac{2}{3}$$

d) ii) $\text{Area} = 41.16 \text{ units}^2$

b) $m_{AB} = \frac{4-0}{8-2} = \frac{2}{3}$

c) $m_1 = m_2$ parallel

$$m_1 = \frac{2}{3} \quad (4, 6)$$

$$y - 6 = \frac{2}{3}(x - 4)$$

$$3y - 18 = 2x - 8$$

$$0 = 2x - 3y + 10$$

f) $d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$

$$a = 2, b = -3, c = 10, x = 2, y =$$

$$d = \frac{|2 \times 2 - 3 \times 0 + 10|}{\sqrt{2^2 + (-3)^2}}$$

$$d = \frac{14}{\sqrt{13}} \text{ or } \frac{14\sqrt{13}}{13}$$

d) $D(-2, 2)$
and plot on diagram

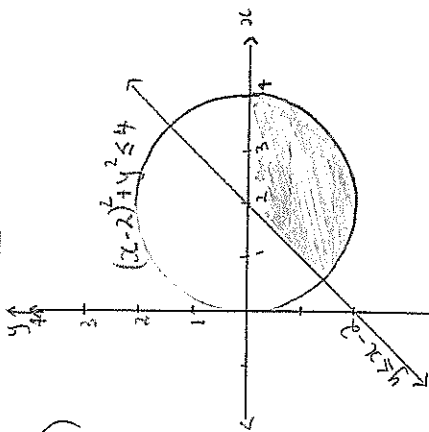
e) $\tan \theta = m$

$$\tan \theta = \frac{2}{3}$$

$$\theta = 34^\circ$$

Question 18.

a)



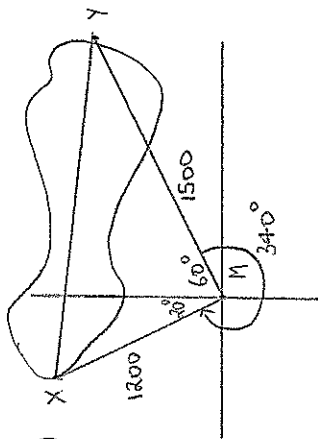
$$b) \sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$$

$$\text{L.H.S} = \sec \theta + \tan \theta = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 + \sin \theta}{\cos \theta}$$

$$= \text{R.H.S}$$

c) i)



$$\text{cii)} \quad c^2 = a^2 + b^2 - 2ab \cos A$$

$$(xy)^2 = 1200^2 + 1500^2 - 2 \times 1200 \times 1500 \times \cos 80^\circ$$

$$xy = 1750.68 \text{ m}$$

$$\text{ciii)} \quad \frac{\sin \theta}{1500} = \frac{\sin 60^\circ}{1750.68}$$

$$\theta = 57^\circ 33'$$

$$= 58^\circ$$

$$180^\circ - (20^\circ + 58^\circ)$$

$$= 102^\circ$$

Bearing of Y from X is 102° T .